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Different micro-aeration rates facilitate production of different end-products from source-diverted blackwater

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ABSTRACT

The effects of micro-aeration on the performance of anaerobic sequencing batch reactors (ASBR) for blackwater treatment were investigated in this study. Different micro-aeration rates, 0, 5, 10, 50, and 150 mg O_2/L-reactor/cycle, and their effect on the hydrolysis, acidogenesis, and methanogenesis of blackwater were evaluated and compared at ambient temperature. Source-diverted blackwater (toilet water) contains high organic contents which can be recovered as biogas. Previous studies have found that anaerobic digestion of blackwater without micro-aeration can only recover upwards of less than 40% of chemical oxygen demand (COD) to methane at room temperature due to the low hydrolysis rate of biomass content in blackwater. This study achieved increases in blackwater hydrolysis (from 34.7% to 48.7%) and methane production (from 39.6% to 50.7%) with controlled micro-aeration (5 mg O_2/L-reactor/cycle). The microbial analysis results showed that hydrolytic/fermentative bacteria and aceto-clastic methanogens (*e.g. Methanosaeta*) were in higher abundances in low-dose micro-aeration reactors (5 and 10 mg O_2/L-reactor/cycle), which facilitated syntrophic interactions between microorganisms. The relative abundance of oxygen-tolerant methanogen such as *Methanosarcina* greatly increased (from 1.5% to 11.4%) after oxygen injection. High oxygen dosages (50 and 150 mg O_2/L-reactor/cycle) led to reduced methane production and higher accumulation of volatile fatty acids, largely due to the oxygen inhibition on methanogens and degradation of organic matters by aerobic growth and respiration, as indicated by the predicted metagenome functions. By combining reactor performance results and microbial community analyses, this study demonstrated that low-dose micro-aeration improves blackwater biomethane recovery by enhancing hydrolysis efficiency and promoting the development of a functional microbial population, while medium to high-dose micro-aeration reduced the activities of certain anaerobes. It was also observed that medium-dose micro-aeration maximizes VFA accumulation, which may be used in two-stage anaerobic digesters.

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1. Introduction

Anaerobic digestion (AD) allows for the recovery of valuable compounds such as biogas and volatile fatty acids from organic wastes. AD has been widely employed for the treatment of various wastes such as agriculture-wastes, food-wastes, and high-strength wastewater (Nag et al., 2019; Panigrahi et al., 2020; Zhang et al., 2019). However, the application of AD for municipal wastewater streams is limited by the low methanogenesis rates due to low organic concentrations and low temperatures (Huang et al., 2019). Recent studies applied AD for source-diverted blackwater (*i.e.*, toilet wastewater) treatment which takes advantage of the high organic content of blackwater in the absence of other domestic wastewater streams, *i.e.*, greywater (Florentino et al., 2019; Zhou et al., 2019). Thermophilic (50–60 °C) and mesophilic temperatures (30–40 °C) have been shown to be suitable for blackwater AD (Gao et al., 2019a; Zhang et al., 2019), while at ambient temperatures, challenges still remain with regards to the chemical oxygen demand (COD) removal efficiency and methane production (Oarga Mulec et al., 2016; Pedrouso et al., 2020).

Hydrolysis is considered as the rate-limiting step in the blackwater treatment process since the blackwater biomass-content is high (25–54 wt% of total solids) (Gao et al., 2019b; Rose et al., 2015). Decomposing of biomass from wastewater can be challenging, especially when operating temperature is below 25 °C, and can

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result in longer hydraulic retention times (HRT) and a larger physical footprint. Various pretreatment technologies (*i.e.*, chemical, mechanical, thermal treatments) have been previously evaluated to improve the hydrolysis of refractory wastes such as lignocellulosic biomass, sludge in wastewater treatment plants and organic solid wastes. Chemicals addition (*e.g.* NaOH, H_2O_2, ozonation) is effective for enhancing the biodegradation of complex materials (Song et al., 2014; You et al., 2019), but can produce inhibitory compounds towards microorganisms, and may cause corrosion of the reaction vessels (Lee and Jeffries, 2011). Mechanical methods, *e.g.* milling and ultrasounds, can be used to improve the solubilization of extra- and intra-cellular organics (Kor-Bicakci and Eskicioglu, 2019; Zheng et al., 2013). Thermal pretreatment has also been shown to be effective in improving substrate hydrolysis, and the heating energy can be compensated with the combustion of produced methane (Toutian et al., 2020). It can be noted that pretreatments often require high operational costs and intensive energy input, energy and cost-efficient technologies are still in demand to facilitate AD performance when treating blackwater. Introducing a small amount of oxygen (*e.g.* micro-aeration) was recently proven to be beneficial to the AD treatment by promoting hydrolysis rates (Duarte et al., 2018; Fu et al., 2016), and was considered as a promising environment-friendly pretreatment strategy to enhance methane recovery from recalcitrant wastes (Nguyen and Khanal, 2018). Oxygen injection can also help to reduce the accumulation of volatile fatty acids (VFA), especially when the loading rate is high (Nguyen et al., 2019). Further, micro-aeration treatment can reduce levels of toxic H_2S from the biogas, which is a more cost-effective (62% cost reduction) approach as compared to chemical absorption (Ramos et al., 2014). Micro-aeration has been applied to different kinds of feedstocks including highly recalcitrant feedstock such as corn straw (Fu et al., 2015), primary sludge (Diak et al., 2013), and co-digestion of brown-water and food-waste (Lim and Wang, 2013), where hydrolysis is critical to solubilize complex organics for downstream conversion. To our best knowledge, no studies to date have investigated the application of micro-aeration for low-temperature anaerobic treatment of blackwater.

Previously oxygen-based applications were often avoided during AD operation because of their potential inhibitory effects on obligate anaerobes. However, recent studies suggested that some strict anaerobes can also adapt to microaerophilic conditions by producing anti-oxidative enzymes or co-existing with facultative microorganisms quickly scavenging oxygen (Angle et al., 2017; Botheju and Bakke, 2011). The balance between these syntrophic microorganisms needs to be maintained by precise oxygen dosage control to prevent over aeration that exceeds the anti-oxidation capacities in a defined system. The effects of micro-aeration vary with regards to enhancing hydrolysis, controlling VFA accumulation, removing hydrogen sulfide, and maintaining suitable conditions for methanogenesis, depending on the amount of oxygen input (Deng et al., 2009; Duarte et al., 2018; Lim et al., 2014). Previous studies have primarily focused on the comparison between anaerobic and microaerobic conditions (Ruan et al., 2019; Yin et al., 2016). Many of these studies were conducted in batch reactor mode (less than 60 days), unable to evaluate the impact of micro-aeration on the AD process in long-term operation, especially with respect to the microbial community structures after long-term acclimation (Nguyen and Khanal, 2018).

This study aims to illustrate the role of micro-aeration pretreatment on the anaerobic digestion of blackwater. The effect of different microaerophilic conditions on the hydrolysis, acidogenesis, and methanogenesis steps were studied. Microbial networks were also investigated to reveal the fundamental impacts of long term continuously oxygen input on AD performance.

2. Materials and methods

2.1. Blackwater collection and characterization

Blackwater (toilet wastewater) was collected from conventional toilets (9 L of water per flush) from the University of Alberta Campus. Influent and effluent samples were filtered through 0.45 μm nylon membrane filters (Fisher Scientific, CA) for soluble component analyses. Total COD and soluble COD (after filtration) concentrations, total solids (TS) and volatile solids (VS) were measured according to the Standard Methods of the American Public Health Association (Rand et al., 1976). Acetate, propionate, and butyrate, being the primary VFAs, were analyzed on a DIONEX ICS-2100 Ionic chromatography (IC) system (Thermo Fisher Scientific, Waltham MA, USA) equipped with a conductivity detector and IonPac AS18 Analytical Column (2 × 250 mm). Samples were diluted and filtered through 0.2 μm nylon membrane filters before VFA analysis.

2.2. Experimental setup and inoculum

Five semi-continuous anaerobic sequencing batch reactors (ASBR) were made using glass with a diameter of 101 mm and a height of 225 mm and a polypropylene cap (inner neck diameter 30 mm). The cap had three ports that were fixed with two longer silicone rubber tubing in the liquid phase and one short silicone rubber tubing in the gas phase. One of the tubes was connected to a gas diffuser submerged in the liquid phase for aeration, the second was used to take liquid samples, and the third for pressure detection and gas sampling. All ports were fixed with plastic valves at the upper end.

The semi-continuous ASBRs were inoculated with 0.5 g VS/reactor (100 mL/reactor) of digested sludge and 0.5 g VS/reactor (130 mL/reactor) of waste activated sludge to ensure both anaerobes and aerobes were present in the system. Reactors were operated at room temperature (22 °C). All reactors were placed in a shaker at 120 rpm. There were five parallel reactors, including AnR, 5-MaR, 10-MaR, 50-MaR and 150-MaR, representing the anaerobic reactor (AnR with no O_2 treatment) and micro-aerobic reactors with gradient aeration levels (MaR with 5, 10, 50, and 150 mg O_2/L-reactor/cycle, respectively). The reactors were operated for 100 days, during which time the performance was evaluated at four different HRTs (5, 4, 3, and 2 days, corresponding to Stages I − IV, respectively). Each cycle consisted of the following steps: filling (5 min, including feeding and air injection), reaction (from 2 to 5 days), settling (30 min) and discharging (5 min). The gas composition analysis was conducted at the end of settling. The exchange ratio was 70% of the working volume.

2.3. Analysis methods

The oxidation reduction potential (ORP) level was measured in the reactors right after air injection to determine the tendency of the aqueous solution in the reactors to accept or donate electrons. ORP can reflect small changes in oxygen concentration which are below the detection limit of dissolved oxygen (DO) probes (0.1 mg/L, which corresponds to −50 mV), and was therefore used in place of DO. ORP was measured by a portable ORP meter (Milwaukee MW500). The pH was measured with a pH meter (B40PCID, VWR, SympHony). Methane production was assessed by measuring the headspace pressure using a pressure meter (GHM 3151, Germany) and the composition of biogas using gas chromatography (GC) (7890B Agilent Technologies, USA). For each gas detection, biogas in reactor headspace was mixed and then withdrawn with a gas-tight syringe. T-test was applied to determine the significance level using

Microsoft Excel® software. A p-value smaller than 0.05 is considered as statistically significant difference.

2.4. Microbial analysis

DNA extraction was performed for the inoculum sludge and sludge in all ASBRs at the end of the experiment using the DNeasy PowerSoil Kit (QIAGEN, Hilden, Germany), according to the manufacturer's protocol. Sludge was sampled three times in one cycle, well mixed, and then centrifuged at 4000 *g* for 10 min to obtain representative sludge samples (Rosenkranz et al., 2013). The pellet was used for DNA extraction. The extracted DNA concentration and quality were examined by NanoDrop One (ThermoFisher, Waltham MA, USA) and gel electrophoresis. PCR was performed followed by sequencing on the Illumina MiSeq platform. The 16S rRNA genes were amplified using the universal primer pair 515F (GTGCCAGCMGCCGCGG) and 806R (GGACTACHVGGGTWTCTAAT). Sequence analysis was performed using the Qiime2 DADA2 pipeline (Callahan et al., 2016; Caporaso et al., 2010) with 99% similarity with reference to the Greengenes database version 13_8 (McDonald et al., 2012; Werner et al., 2012). Alpha and Beta diversities and principal coordinates analysis (PCoA) using Bray-Curtis distance were performed using the "vegan" package (Jari Oksanen et al., 2017) in RStudio version 3.4.1. The predicted metagenome and functional genes were determined using the Phylogenetic Investigation of Communities by Reconstruction of Unobserved States (PICRUSt) (Langille et al., 2013). Correlation was computed using the R "corrplot" package using Pearson's product-moment correlation test (Wei and Simko, 2017).

3. Results

3.1. Blackwater characterization and reactor performances

The characteristics of blackwater from toilets on campus varied slightly in every collection. The total chemical oxygen demand (TCOD) of the blackwater was 1000 ± 70 mg/L and the corresponding soluble chemical oxygen demand (SCOD) of the fresh blackwater was 296 ± 74 mg/L, accounting for 20–30% of TCOD. The blackwater pH varied from 7.2 to 8.0, depending on the storage time and the solubilization of the feedstock. The total ammonia nitrogen (TAN), total nitrogen (TN) and total phosphorus (TP) concentrations are shown in Table S1. The five ASBRs operated with different levels of air injection were operated for 100 days, under four different HRT conditions (initially set as 5 days then gradually shortened 1 day at a time to 2 days).

The impact of micro-aeration treatment on hydrolysis is shown in Fig. 1A. Despite the relatively large standard deviation observed due to the fluctuate blackwater quality, statistical analysis results demonstrated the impacts of micro-aeration on blackwater digestibility ($P < 0.05$). For all HRT conditions, the low-dose micro-aeration in 5-MaR (5 mg O_2/L-reactor/cycle) resulted in the highest hydrolysis rates when compared to the no micro-aeration condition (AnR), and micro-aeration at higher dosages (10-MaR, 50-MaR, and 150-MaR); this indicates that low-dose micro-aeration can enhance hydrolysis efficiency. As the micro-aeration dose increased, there was a decreasing trend in the hydrolysis rate. In particular, under Stage IV (HRT of 2 days), the 5-MaR and 10-MaR reactors had significantly higher ($P < 0.01$) hydrolysis efficiencies (48.7 ± 3.3% for 5-MaR and 39.9 ± 5.8% for 10-MaR), as compared to the AnR (34.7 ± 3.7%). Under this condition, the 50-MaR exhibited a comparable hydrolysis efficiency (37.7 ± 5.6%), as compared to AnR ($P < 0.01$), while higher micro-aeration doses (*i.e.*, 150-MaR with 25.1 ± 6.5%) led to a significant reduction in hydrolysis efficiency ($P < 0.01$).

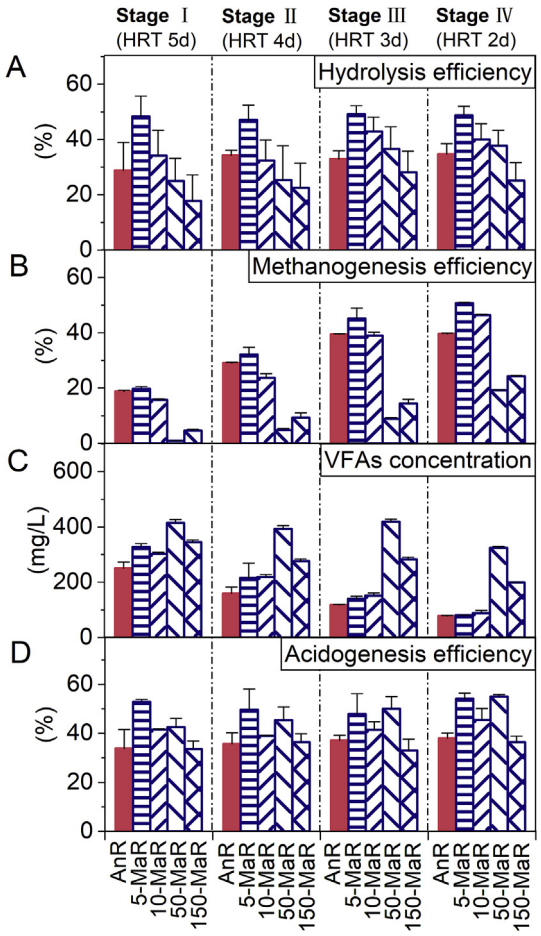


Fig. 1. Reactor performances including (A) hydrolysis efficiency, (B) methanogenesis efficiency, (C) total VFAs production, and (D) acidogenesis efficiency. Error bars represent standard deviation calculated from the stabilized cycles in each stage.

The stabilized methanogenesis efficiency in each stage of the operation is shown in Fig. 1B. Reactor ORP values are provided in Fig. S1B. For all five reactors, the methanogenesis rate increased with a reduction in HRT. This observation may be attributed to the establishment of more efficient methanogen communities at higher organic loading rates, similar to results reported previously (Lim and Wang, 2013; Ramos and Fdz-Polanco, 2013). At the end of Stage IV, the methanogenesis efficiency reached 50.5% and 46.1% in 5-MaR and 10-MaR, respectively, higher than that in AnR (39.9%). However, significantly lower methanogenesis efficiencies were observed in the 50-MaR (19.3%) and 150-MaR (24.6%) reactors, likely due to the inhibitory effect of increased oxygen dosage to methanogens. Although the effluent ORP of all five reactors was maintained below −300 mV under all conditions, the reduced methanogenesis rates at higher micro-aeration dosages indicate relatively higher ORP (as shown in Fig. S1B) had a considerable impact on methanogen activities.

The effects of micro-aeration intensity on effluent VFA concentrations are shown in Fig. 1C. Detailed information on VFA compositions and reactor pH changes are provided in Figs. S2 and S1A. As shown in Fig. 1C, effluent VFA concentrations exhibited decreasing trends as the HRT was reduced from Stage I to IV, correlating with the increasing methanogenesis efficiencies discussed above. The 50-MaR condition showed the highest VFA accumulatio in all stages, also correlating to the lowest methanogenesis activities in 50-MaR (Fig. 1B), possibly due to the VFA

inhibition of methanogens. In Stage IV, 50-MaR had the highest VFA concentration (324.5 ± 4.5 mg/L), which was more than three times higher than that observed in AnR, 5-MaR, and 10-MaR, and was slightly higher than the VFAs produced from 150-MaR (199 ± 1 mg/L). As compared to the 150-MaR, 50-MaR had higher hydrolysis efficiency, but lower methanogenesis efficiency, both contributing to the higher VFA accumulation.

The acidogenesis efficiency (accounting for the proportion transformed to methane) is shown in Fig. 1D. The acidogenesis efficiency was significantly higher ($P < 0.01$) in 5-MaR, 10-MaR, and 50-MaR (54.1 ± 2.3%, 45.4 ± 4.7% and 55.0 ± 0.9% in last stage, respectively), as compared to AnR (38.0 ± 2.2%). This indicates that micro-aeration at relatively lower dosages improved acidogenesis. Further, since the different component is mainly acetate (Supplementary Fig. S2), it reveals that the acetogenesis step of anaerobic digestion can also be enhanced.

3.2. Microbial community analysis

The microbial community alpha-diversity and number of different genera, which indicates community richness, is shown in Fig. 2A for bacteria and archaea. Both bacterial and archaeal communities exhibited a decreasing trend with the increasing oxygen dosage, which may be explained by the inhibitory effects of oxygen on some of the obligate anaerobic microorganisms. The PCoA of Bray-Curtis distance (Fig. 2B) shows that bacterial and archaeal communities were clustered into an AnR, 5-MaR, and 10-MaR group, and 50-MaR and 150-MaR group, and shifted away from the inoculum community. The clustering pattern is in close relation to the methanogenesis efficiency pattern (Fig. 1B) where AnR, 5-MaR, and 10-MaR showed similar values while 50-MaR and 150-MaR showed impaired levels of methanogenesis. The total abundance ratios of archaea and bacteria showed a decreasing trend with increasing oxygen dosage (Fig. 2B), which further confirmed that oxygen has an inhibitory effect on methanogens.

The comparison of archaeal genera composition is shown in Fig. 3. The AnR, 5-MaR, and 10-MaR were dominated by *Methanosaeta*, which is an acetoclastic methanogen. The abundance of *Methanosaeta* was remarkably lower at higher micro-aeration dosage conditions (50-MaR and 150-MaR). In comparison, *Methanosarcina*, which can utilize both acetate and H_2/CO_2 to produce methane, was less affected by the oxygen dosage. The decreasing order of the sum of *Methanosaeta* and *Methanosarcina* abundances in 5-MaR, 10-MaR, AnR, 150-MaR, and 50-MaR is well correlated with methanogenesis efficiencies at the final steady state. On the other hand, *Methanospirillum*, a hydrogenotrophic strain, had a higher abundance in the higher oxygen dosage reactors (50-MaR and 150-MaR), indicating that hydrogenotrophic methanogens may have a higher tolerance to oxygen and thus become the dominant methane-production pathway under higher oxygen levels. Further, *f_Methanomassiliicoccaceae* was present in all reactors but also higher in these two reactors (41% in 50-MaR and 35% in 150-MaR), indicating that obligately methylotrophic may proliferate under higher oxygen conditions.

The bacterial composition at the family level of the most abundant 10 families in each reactor is shown in Fig. 3B. 5-MaR and 10-MaR shared similar abundances of bacterial families, while the 50-MaR and 150-MaR shared similar abundances. One unidentified family in the order *Bacteroidales* (*o__Bacteroidales[2]*) and *Desulfomicrobiaceae* were highly enriched in 5-MaR (26% and 17%, respectively) and 10-MaR (18% and 16%, respectively). *Bacteroidales* makes up a significant portion of fecal bacterial population (Ridley et al., 2014) and have been reported to function as hydrolytic/fermentative bacteria of large macromolecules such as cellulose and protein (Azman et al., 2017; Yi et al., 2014) The enrichment of these fermenters and sulfate-reducing bacteria in low micro-aeration intensity reactors may be attributed to enhanced substrate provided by hydrolysis. At higher oxygen dosages, certain families showed increased abundance, including *SB-1* from the phylum *Bacteroidetes*, one unidentified family in the order *SHA-98*

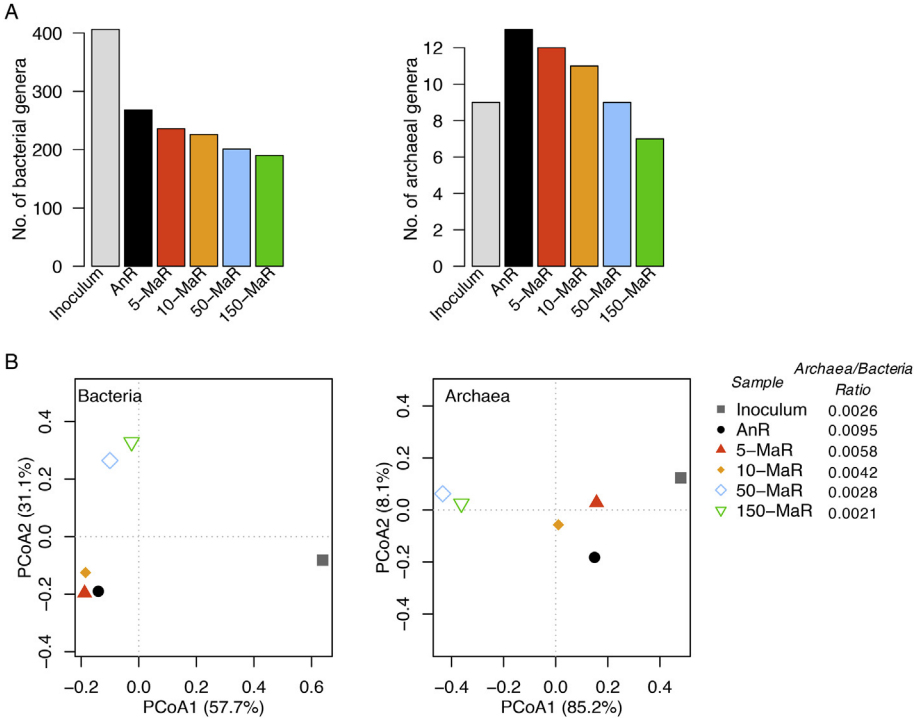


Fig. 2. Microbial community diversity. (A) Community richness indicated by the number of genera. (B) PCoA of Bray-Curtis distance.

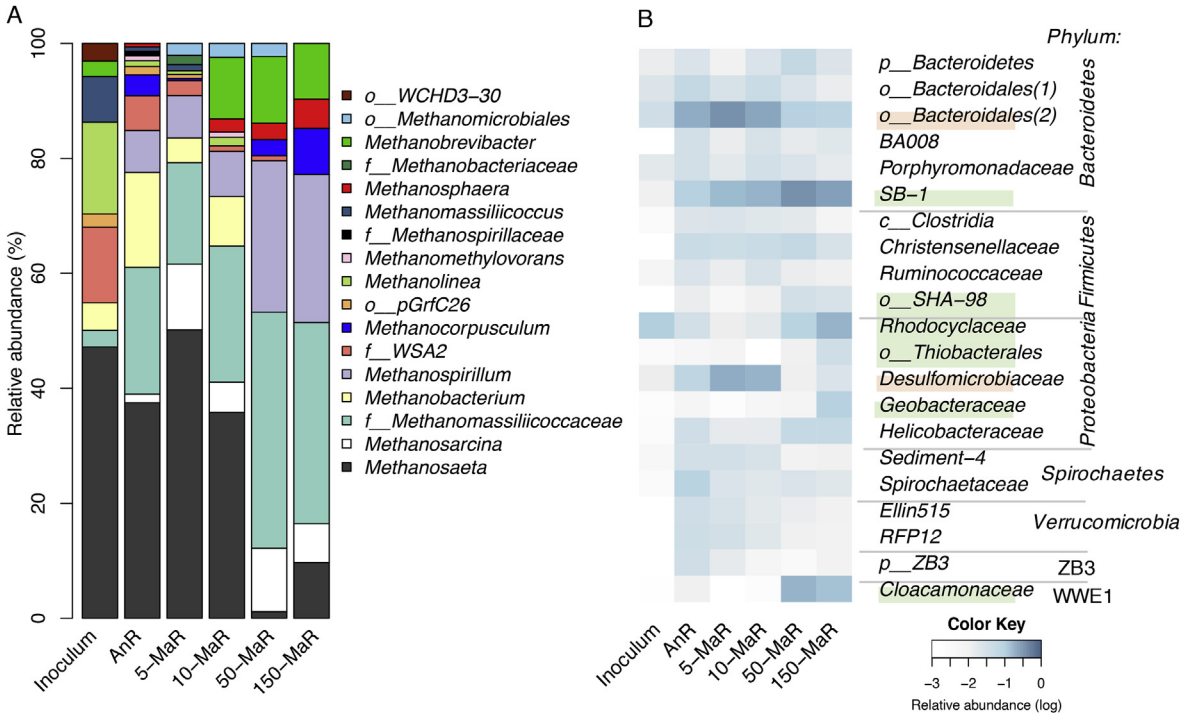


Fig. 3. (A) The relative abundance of archaeal genera (B) The most abundant 10 bacterial families in samples of inoculum, AnR, 5-MaR, 10-MaR, 50-MaR, and 150-MaR. The orange shade indicates higher abundance in 5-MaR, the green shade indicates higher abundance in 50-MaR or 150-MaR. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

from the phylum Firmicutes, *Rhodocyclaceae*, one unidentified family in the order *Thiobacterales*, *Geobacteraceae* from the phylum *Proteobacteria*, and *Cloacamonaceae* from the phylum *WWE1*. *Bacteroidetes* is a broad group of microorganisms that mainly ferment polysaccharides or carbohydrates (Braz et al., 2019). 5-MaR, 10-MaR, and 50-MaR all showed higher *Bacteroidetes* levels (45%, 44% and 43%, respectively) compared to AnR (37%), which correlates well with the hydrolysis activities observed. The increase of *Cloacamonaceae* may be related to the VFA accumulation in 50-MaR and 150-MaR since these microorganisms may ferment amino acids and syntrophically oxidize propionate (Braz et al., 2019; Goux et al., 2015). Decreasing trends in abundance with increasing oxygen dosage were observed in the families *Sediment-4* and *Spirochaetaceae* (phylum *Spirochaetes*), *Ellin515* and *RFP12* (phylum *Verrucomicrobia*), and one unidentified family in the phylum *ZB3*, indicating their inhibited activity under high oxygen dosage.

The metagenome and functional genes were predicted from the 16S rRNA gene amplicon data using a reference database (Langille et al., 2013). Different oxygen dosages promoted different functional groups as shown in Fig. 4. In the metabolism function group, 50-MaR showed the highest prevalence of the biosynthesis and metabolism of glycan, and the metabolism of carbohydrates, nucleotides, terpenoids, polyketides, cofactors, vitamins, and lipids, which can be associated with the high VFA production and accumulation in 50-MaR. The 50-MaR reactor also exhibited the highest prevalence of genetic information processing functions of replication and repair, translation, folding, sorting, and degradation, which could be triggered by the acidic environment resulting from VFA accumulation. The highest oxygen dosage (150-MaR) improved functions of xenobiotics biodegradation and metabolism as well as signal transduction and membrane transport, which require sufficient energy and maybe promoted at higher ORP (Fig. S1B). Moreover, cell mobility, growth, and death functional genes were the highest in 150-MaR, indicating active growth of cells, which may have resulted from the promoted activity of aerobic microorganisms.

3.3. Correlation between parameters

Fig. 5 depicts the Pearson's product-moment correlation matrix between oxygen dosage and reactor performance (effluent VFAs, effluent SCOD, methanogenesis, hydrolysis, and acidogenesis), pH and archaea/bacteria ratio, and the significance level ($P < 0.05$) of the correlation. Oxygen dosage had a significant negative correlation with the hydrolysis and acidogenesis efficiencies. Further, higher oxygen dosages were significantly negatively correlated with methanogenesis, pH, and archaea ratio, resulting in higher effluent VFAs and SCOD since the methanogenesis was inhibited and organic substrates were accumulated. Fig. 5 shows a significant positive correlation between effluent VFAs and SCOD since the SCOD was primarily composed of fermented substrates. Methanogenesis was significantly negatively correlated with effluent VFAs and SCOD, because the VFAs and SCOD were either transformed to methane or accumulated in the reactor. A positive correlation cluster between methanogenesis, hydrolysis, acidogenesis, pH and archaea ratio is shown in the lower part of Fig. 5, indicating that the biochemical reactions in anaerobic digestion process are interrelated; this emphasizes the importance of maintaining an appropriate pH and archaea ratio.

4. Discussion

4.1. Balance between anaerobic and aerobic activities

Low oxygen dosages could promote hydrolysis and acidogenesis bacterial activity. The growth of *Bacteroidetes* was enhanced at lower oxygen dosages compared to anaerobic conditions (Fig. 3B). Large molecules were broken down into smaller molecules such as

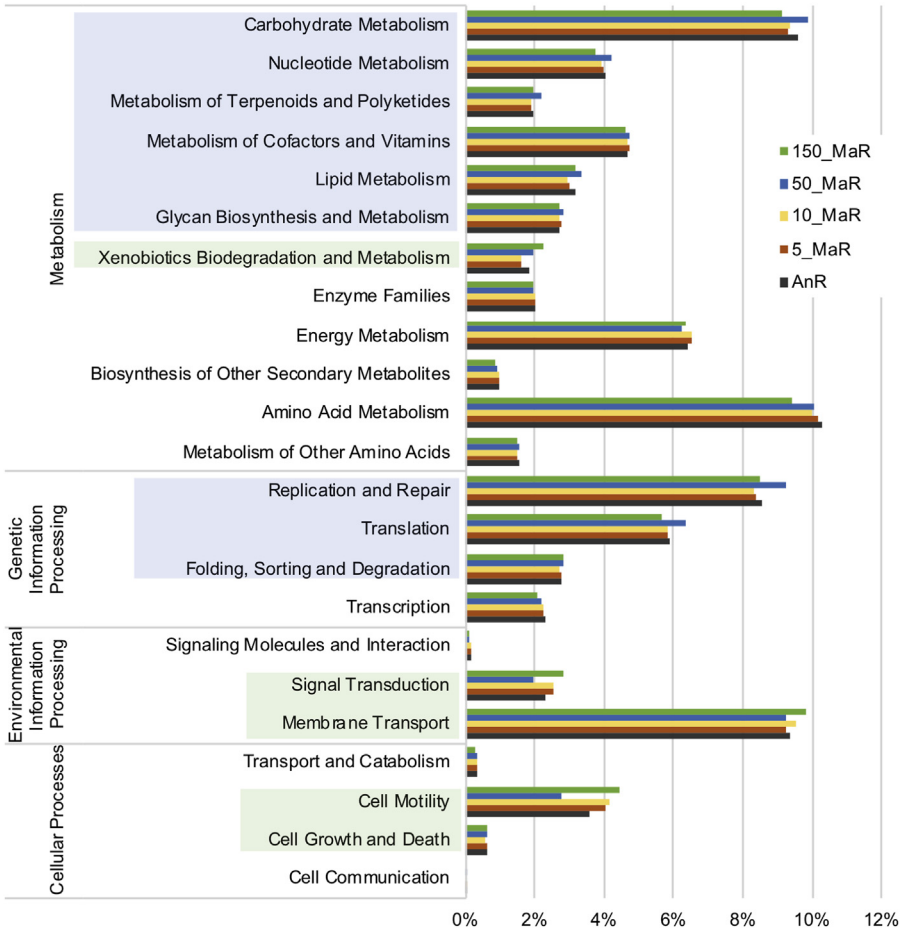


Fig. 4. Predicated metagenome functions. The blue shade indicates a higher abundance in 50-MaR, and the green shade indicates a higher abundance in 150-MaR. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

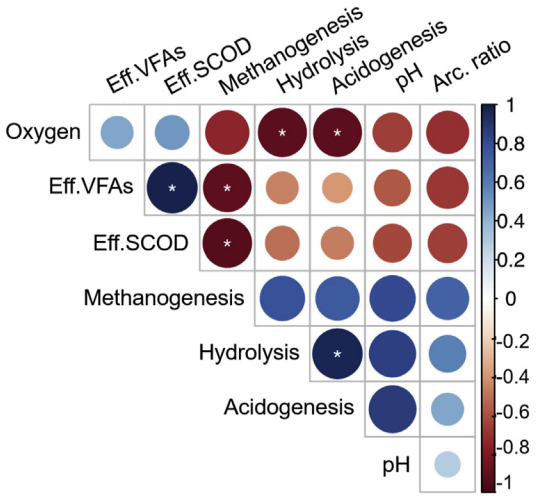


Fig. 5. Correlogram of reactor parameters and archaea ratio in the total community (star denotes significance $P < 0.05$).

tyrosine (Fig. S3). By manipulating the oxygen level in the extracellular environment, intracellular metabolic pathways and oxidation related gene expressions of microbes can also be changed (Liu et al., 2013). The predicted metagenome functions of medium oxygen dosages (Fig. 4) exhibited higher functional genes of large-molecule metabolism (carbohydrate, nucleotide, terpenoids, polyketides, cofactors, vitamins, lipids, and glycan). However, too much oxygen may promote aerobic microorganism growth, *e.g. Rhodocyclaceae* grow under high oxygen dosages (Fig. 3B), and inhibit methanogens, especially *Methanosaeta* (Fig. 3A). In micro-aerobic environment, anti-oxidative enzymes are produced to protect cells. *Bacteroides fragilis* (*p__Bacteroidetes*), *Desulfovibrio gigas* (*p__Proteobacteria*), and *Methanosarcina barkeri* (*g__Methanosarcina*) were reported to show 2 to 30 times overexpression of genes encoding anti-oxidative enzymes under an oxidative stress as compared to anaerobic conditions (Brioukhanov and Netrusov, 2007; Brioukhanov et al., 2006).

4.2. End product-oriented oxygen dosage control

The micro-aeration intensity varies depending on specific purpose and feed composition. There are four interdependent steps during anaerobic degradation of organic waste, and the products excreted by one step serve as the substrates for another group of microbes and thus should be kept kinetically constant. To avoid the inhibition of methanogens and harvest the highest methane yield, low oxygen dosages should be applied when designing a micro-aeration-based AD system. However, the hydrolytic acidification process is also considered an economical and effective pretreatment method, especially for wastewater containing highly recalcitrant and complex pollutants (Braz et al., 2019; Tsapekos et al., 2017). A medium oxygen dosage (50 mg O_2/L-reactor/cycle, 5% of

influent COD) is the threshold oxygen level for methanogen inhibition, cutting off the methanogenesis step for VFA production (Duan et al., 2017; Xu et al., 2014). The syntrophic anaerobic oxidation of VFAs to acetate and H_2 is inhibited under medium micro-aerobic condition, as indicated by the lowest *Proteobacteria* level (13%), which include glucose and VFAs utilizing bacteria (Ariesyady et al., 2007). Further, higher oxygen level (150 mg O_2/L-reactor/cycle) is sufficient to facilitate the highly energetic metabolism of aerobic VFAs oxidation by heterotrophs (Zhu et al., 2009). Previous studies showed that in order to enhance hydrolysis, micro-aeration rates can be calculated based on 2–10% of substrate COD (Botheju and Bakke, 2011), while in our study a much lower oxygen dosage (0.5% of influent COD) was needed to boost hydrolysis, offering a cost-efficient strategy.

5. Conclusion

This study evaluated the effects of five micro-aeration intensities (0, 5, 10, 50, and 150 mg O_2/L-reactor/cycle) on the hydrolysis, acidogenesis, and methane yield of blackwater in five ASBRs. The optimum micro-aeration intensity was 5 mg O_2/L-reactor/cycle (5-MaR), which accelerated COD solubilization (hydrolysis efficiency 48.7 ± 3.33%) and enhanced methane production (methanogenesis efficiency 50.7 ± 0.20%) by 40.3% and 28.1%, respectively, when compared to no aeration (hydrolysis efficiency 34.7 ± 3.75% and methanogenesis efficiency 39.6 ± 0.28%). However, excessive oxygen (150 mg O_2/L-reactor/cycle) had a negative effect on the AD process. Aerobic respiration consumes some organics which were supposed to be recovered as energy. Further, strict anaerobes were inhibited before oxygen was consumed. VFAs accumulated in the 50 mg O_2/L-reactor/cycle reactor (ranging from 324.5 mg/L to 418.3 mg/L) due to the lack of downstream VFA conversion, so medium-dose micro-aeration offers an operational strategy for a two-stage anaerobic digestion system. Oxygen impaired the diversity of bacteria and archaea but enriched more competitive microorganisms which can recover higher methane compared to the fully anaerobic reactor.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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